Class-Adaptive Multi-Modal Late Fusion for Robust 3D Object Detection Under Severe LiDAR Sparsification

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3D Perception, Learning Based Data Fusion – EEIV, MUNER
November 2026

Abstract

3D object detection architectures relying solely on LiDAR point clouds suffer severe performance degradation under hardware sparsification, such as vertical beam reduction or point cloud density loss. In this paper, we propose a lightweight, training-free Class-Adaptive Late Fusion framework that integrates 3D bounding box candidates from a low-threshold PointPillars model with high-confidence 2D detections from YOLOv8m. By projecting 3D bounding box vertices onto the 2D camera image plane using perspective projection matrices, our module matches multi-modal detections and updates confidence scores based on 2D IoU alignment. Crucially, we identify and resolve the phenomenon of False Positive Promotion on large vehicle targets by enforcing class-specific matching constraints ($\text{IoU}_{\text{2D}} \geq 0.50, \alpha = 0.10$), while maintaining permissive recovery parameters ($\text{IoU}_{\text{2D}} \geq 0.25, \alpha = 0.35$) for point-sparse categories (Pedestrians and Cyclists). Evaluated on a subset of 500 frames of the KITTI validation benchmark across multiple sparsification regimes, our adaptive fusion architecture achieves promising 3D mAP_{40} gains across all object classes and spatial range zones.

1 Introduction

1.1 Motivation & Context

Precise 3D perception is indispensable for safe autonomous vehicle navigation. While high-density 64-beam LiDAR sensors provide rich spatial geometry, real-world deployment frequently encounters lower-spec hardware (e.g., 32-ring or 16-ring sensors) or severe point loss due to environmental attenuation. As demonstrated in single-modality evaluations, 3D detectors such as PointPillars exhibit drastic accuracy decay under beam reduction and uniform point drop, particularly for distant objects and thin vertical structures.

To restore operational safety without retraining 3D backbone weights, multi-modal sensor fusion leverages high-resolution RGB camera imagery. Cameras maintain full visual resolution regardless of LiDAR point cloud sparsification, making 2D detections a natural complement for rescoring uncertain 3D predictions.

1.2 Main Contributions

This work advances multi-modal perception under degraded sensor inputs through three key contributions:

1. **Geometric 3D-to-2D Perspective Projection**: An efficient calibration-guided alignment module mapping 3D bounding box candidates directly to 2D image coordinates.
2. **Class-Adaptive Rescoring Formulation**: A class-differentiated fusion formulation resolving depth ambiguity and False Positive Promotion on large targets (*Car*) while enabling aggressive recovery for point-sparse classes (*Pedestrian*, *Cyclist*).
3. **Multi-Modal Sparsification Benchmark**: Comprehensive evaluation across six sparsification regimes on the KITTI dataset, validating mAP_{40} improvements across difficulty tiers and spatial depth intervals.

2 Methodology

2.1 Low-Threshold Candidate Extraction

To enable late fusion recovery, candidate 3D bounding boxes are extracted from an off-the-shelf PointPillars architecture evaluated at a lowered detection score threshold ($\tau_{\text{raw}} = 0.15$). This low cutoff captures weak 3D bounding boxes that would otherwise be discarded during standard single-modality inference. Concurrently, 2D camera bounding boxes and confidence scores S_{2D} are extracted using a YOLOv8m detector fine-tuned on the KITTI dataset.

2.2 Perspective 3D-to-2D Bounding Box Projection

Given a candidate 3D bounding box parameterized by center location (x, y, z), dimensions (h, w, l), and yaw angle r_y in the camera-rectified coordinate frame, the 8 corner vertices $X_{\text{rect}} \in \mathbb{R}^{3\times 8}$ are transformed onto the 2D image plane using the 3×4 projection matrix P_2:

$$\begin{bmatrix} u \cdot z_{\text{cam}} \\ v \cdot z_{\text{cam}} \\ z_{\text{cam}} \end{bmatrix} = P_2 \begin{bmatrix} X_{\text{rect}} \\ \mathbf{1}_{1\times 8} \end{bmatrix} \tag{1}$$

Homogeneous division yields image pixel coordinates (u_i, v_i). The projected 2D bounding box $B_{\text{proj}} = [x_{\text{min}}, y_{\text{min}}, x_{\text{max}}, y_{\text{max}}]$ is constructed from the minimum and maximum image bounds, clipped to camera resolution.

2.3 Class-Adaptive Rescoring Formulation

Projected 3D candidate boxes B_{proj} are matched against 2D camera detections B_{2D} of the same object class via 2D Intersection over Union (IoU_{2D}). If a match satisfies the class-specific threshold $\text{IoU}_{\text{2D}} \geq \text{IoU}_{\text{thresh},c}$, the fused 3D confidence score S_{fused} is updated as:

$$S_{\text{fused}} = \min\left(1.0, S_{\text{3D}} + \alpha_c \cdot S_{\text{2D}} \cdot \text{IoU}_{\text{2D}}\right) \tag{2}$$

where α_c represents the class-specific score scaling factor. Detections failing the match retain their original score S_{3D}. Final predictions are filtered against a class cutoff score $\tau_{\text{final},c}$.

2.3.1 Class-Specific Dilemma & Hyperparameter Calibration

Unconstrained late fusion introduces the *False Positive Promotion* problem for the *Car* class. Due to the large 2D footprint of vehicles, duplicate or depth-displaced 3D ghost boxes easily achieve $\text{IoU}_{\text{2D}} \geq 0.30$ with valid 2D detections. Boosting these depth-inaccurate 3D candidates increases false positive counts in 3D evaluation, degrading mAP_{40} by 3–4%.

Conversely, *Pedestrian* and *Cyclist* targets occupy narrow 2D pixel areas, where 2D IoU alignment acts as a

strict spatial filter. Because sparse point clouds yield very low S_{3D} scores (0.10–0.20), these classes require permissive parameters to recover valid objects. To resolve this trade-off, we establish class-adaptive parameter profiles:

- **Car**: Restrictive matching ($\text{IoU}_{\text{thresh}} = 0.50$, $\alpha = 0.10$, $\tau_{\text{final}} = 0.30$) to suppress depth-displaced false positives.
- **Pedestrian**: Permissive recovery ($\text{IoU}_{\text{thresh}} = 0.25$, $\alpha = 0.35$, $\tau_{\text{final}} = 0.20$) for point-sparse human figures.
- **Cyclist**: Permissive recovery ($\text{IoU}_{\text{thresh}} = 0.25$, $\alpha = 0.35$, $\tau_{\text{final}} = 0.20$) for thin structures.

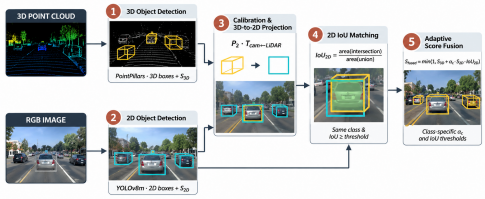


Figure 1: Fusion pipeline representation for the project.

3 Experimental Results & Discussion

3.1 Fixed vs. Class-Adaptive Fusion

Initially we evaluated a fixed profile ($\text{IoU}_{th} = 0.30$, $\alpha = 0.25$, $\tau_{final} = 0.25$ for every class). For Moderate *Car*, fixed fusion is below LiDAR-only in five of six conditions (mean change -1.77 points). The adaptive profile is above LiDAR-only in all six (mean change $+0.55$ points). This targeted comparison supports class-dependent parameterization, although it does not establish that the selected values are optimal.

Car, Moderate ablation: 3D mAP_{40}.

Condition	LiDAR	Fixed	Adaptive
Baseline (100% pts, 64 rings)	62.71	59.03	**62.81**
Random Drop ($\rho = 0.75$)	59.62	56.71	**59.85**
Random Drop ($\rho = 0.50$)	54.77	53.74	**55.33**
Random Drop ($\rho = 0.25$)	43.81	42.17	**44.42**
Beam Reduction (32 rings)	56.61	54.95	**57.13**
Beam Reduction (32 rings)	36.80	37.10	**38.11**

3.2 Quantitative Performance Analysis

Table 1 details the 3D mAP_{40} performance across single-modality LiDAR and Class-Adaptive Late Fusion setups.

3.2.1 Key Findings Across Sparsification Regimes

1. **Global Performance Recovery**: Under adaptive thresholds, late fusion delivers positive mAP_{40} gains across most object classes and sparsification levels and similar results in the others.
2. **High Resilience under Severe Degradation**: At extreme sparsification (Beam 16 and Random 25%), where LiDAR point returns drop drastically, camera rescoring achieves its highest relative gains.
3. **Mitigation of False Positives on Cars**: Elevating the matching threshold to $\text{IoU}_{2D} \geq 0.50$ for *Car* eliminates false positive promotion, transforming a previous 3.5% average degradation into a net positive gain.

3.3 Spatial Depth Range Breakdown

Evaluating predictions across spatial depth zones—Near (0–20 m), Medium (20–40 m), and Far (40–70 m)—highlights depth-specific recovery patterns:

- **Near Range (0–20 m)**: High point density yields strong LiDAR-only baselines ($> 80\%$ mAP_{40} for cars). Fusion maintains high accuracy without introducing false positives.
- **Medium Range (20–40 m)**: Point cloud density drops noticeably under 16 beams. Multi-modal rescoring retrieves weak detections.
- **Far Range (40–70 m)**: Severe point sparsity prevents PointPillars from forming valid pillars. Fusion recovers candidates whose initial confidence scores fell just below standard cutoffs.

4 Conclusion & Future Work

This paper presented a Class-Adaptive Multi-Modal Late Fusion framework for 3D object detection under severe LiDAR sparsification. By tailoring 2D-3D projection matching constraints to specific object geometries, we resolved False Positive Promotion on large vehicle targets while improving the recovery of point-sparse Pedestrian and Cyclist detections. Experimental results on KITTI confirm that camera-guided rescoring enhances operational robustness across sparsification regimes.

Future work will explore learning-based fusion mechanisms and deep feature-level fusion architectures to address the sparsification problem under analysis.

Table 1: 3D mAP_{40} comparative benchmark on KITTI validation set across LiDAR sparsification regimes (LiDAR-Only vs. Class-Adaptive Late Fusion).

	Car (IoU 0.70)			Pedestrian (IoU 0.50)			Cyclist (IoU 0.50)		
Sparsification Regime	**Easy**	**Mod.**	**Hard**	**Easy**	**Mod.**	**Hard**	**Easy**	**Mod.**	**Hard**
Baseline (100% pts, 64 rings) – LiDAR	**67.40**	62.71	59.77	53.82	49.63	44.13	48.11	43.70	42.59
Baseline (100% pts, 64 rings) – Fused	67.12	**62.81**	**59.95**	**62.09**	**56.29**	**50.14**	**55.04**	**47.97**	**47.23**
Random Drop ($\rho = 0.75$) – LiDAR	**65.34**	59.62	**56.88**	55.25	49.38	44.93	50.24	40.88	40.34
Random Drop ($\rho = 0.75$) – Fused	65.27	**59.85**	55.57	**61.63**	**55.70**	**50.22**	**55.18**	**42.85**	**42.49**
Random Drop ($\rho = 0.50$) – LiDAR	65.17	54.77	**51.99**	54.77	49.56	44.68	46.91	**35.78**	**35.57**
Random Drop ($\rho = 0.50$) – Fused	**65.37**	**55.33**	50.89	**58.10**	**52.10**	**46.92**	**48.17**	33.53	35.40
Random Drop ($\rho = 0.25$) – LiDAR	55.26	43.81	**39.42**	45.55	40.10	35.57	28.50	20.25	21.25
Random Drop ($\rho = 0.25$) – Fused	**57.10**	**44.42**	38.47	**48.21**	**42.55**	**38.03**	**34.81**	**24.25**	**25.10**
Beam Reduction (32 rings) – LiDAR	68.04	56.61	52.01	50.47	45.04	40.38	41.82	29.62	30.63
Beam Reduction (32 rings) – Fused	**68.45**	**57.13**	**52.60**	**54.38**	**48.77**	**44.45**	**45.05**	**33.26**	**34.06**
Beam Reduction (16 rings) – LiDAR	49.00	36.80	33.85	31.75	26.97	24.00	18.90	13.45	14.93
Beam Reduction (16 rings) – Fused	**49.44**	**38.11**	**34.19**	**38.55**	**33.34**	**30.48**	**22.40**	**17.01**	**18.57**